Presentation 4

What did each of the team members do?

Note that multiple team members can work together on the same task. In that case, you may write the same task for all of them. For example,

Benjamin Pilnick: I worked on OPTUNA hyperparameter tuning. I conducted 4 separate studies of OPTUNA tuning starting with the original suggested hyperparameters in the file optuna_training.py. Each subsequent trial I changed the hyperparameter suggested parameters ranges based upon the previous trials. Each study had differing amounts of Sobol and TPE trials. Trial one had 16 while trial two had 10, and the rest had 8. Was able to converge on optimal values of learning rate and patch dropout. I found the best model was my round 1 model with a validation loss of 0.225 and a test loss of 0.589. Results exist here: Database results from OPTUNA

Akshay Anand: I created a mathematical argument for the small data memorization issues that we faced. Why did the training loss plummeted to zero while validation loss was sky high? What could be causing the major generalization issues?

Isabella Toth: I worked on both Coding Objective 1 and Coding Objective 2. For Objective 1, I used the results of each of the 5 splits for the round of model tuning that had the best performing hyperparameter configuration to create plots of training and validation loss vs. epoch, confusion matrices, deviation plots, and visualizations of the effective high-attention patches for the misclassified benign and high-grade cases. I explained what each of these visualizations meant in my presentation. For Objective 2, I made a boxplot of the best validation loss obtained across the 10 runs for each split. I also made a boxplot of the predicted probability across the 10 runs within each of the splits for each test case. I explained and interpreted these results.

Zoe Lively: I implemented the new code for objective 2 for the effect of patch order on accuracy (only made changes to main.py and trainer.py). We set 10 random seeds for each of the 5 splits, and then added 4 csv outputs to analyze for the presentation. I then evaluated the results of Classification consistency and accuracy variability. I found that overall cases seem to be consistent, with just 19 cases below 80% correct. The accuracy was generally high across the board, but split 2 and 3 were a bit more sensitive to patch order, while 4 was more stable (less variability). But their midpoints remain consistent showing that the performance overall is pretty reliable. Lastly I evaluated the correlation of classification and val loss, and found that while they seem to have a linear relationship, there isn't anything strong enough at the moment to draw a correlation or causation.

How does it help the project?

OPTUNA tuning: By tuning for optimal parameters we can increase the Benign/High Grade recall, accuracy, and reduce test loss. Comparing what we started with at the beginning of the quarter we have made significant progress in updating the accuracy of the model. It has more stable predictions across the 5 splits and better overall accuracy.

Mathematical Argument: Helps provide possible solutions for the issues and understand what is going on. A model that doesn't generalize well is a poor model and therefore it is best to solve generalization issues early on.

Visualizations/Effect of Patch Order on Accuracy: By visualizing the results, we can assess our model's performance and understand the conceptual shortcomings of our code. In doing so, we can pinpoint specific areas for improvement and streamline our efforts into fixing the areas that require the most refinement. Additionally, using multiple seeds shows how stable and reliable the model really is. Understanding how the model is affected by this helps us create one that performs consistently and is more trustworthy than one with high but unstable accuracy.

Issues faced that haven't been resolved:

Proposed next steps:

- Model Tuning: Further tune OPTUNA_training.py and potentially change the sampler. The TPE sampler had no effect on the optimization history, the lowest validation loss occurred in the SOBOL trials.
- Experiment with and increase different batch sizes to reduce sensitivity to patch order and stabilize training.
- Explore ways/ideas to further stabilize gradient updates for those remaining 19 cases

References